CAPSTONE | BUILD & EVALUATE AN LLM-POWERED APPLICATION

Cross-Market Financial Quant & Filing Analyst Agent

A tool-augmented ReAct agent that computes exact ratios, cites real SEC filings, and abstains when it should

25.2% → 69.5%
Numerical accuracy

92.9%
Citation precision

0% → 80%
Trap abstention

Aungruk Vanichanai | KMITL, School of Information Technology

The Problem

Three bottlenecks make plain LLMs unsafe for filing analysis



Unreliable arithmetic

Next-token prediction can't reliably compute DuPont ROE, Altman Z-Score or FCF CAGR.



Temporal confusion

Standard RAG mixes 9-month cumulative, single-quarter and full-year guidance numbers.



Confident hallucination

Unknown companies or metrics get plausible falsehoods instead of an abstention.

Who It's For

Analysts who need exact numbers with verifiable provenance



Equity research & quant funds

Extract and validate quarterly metrics in seconds instead of hours of spreadsheet work.



Credit & risk officers

Altman Z-Score with SAFE / GREY / DISTRESS zones, linked to filing footnotes.

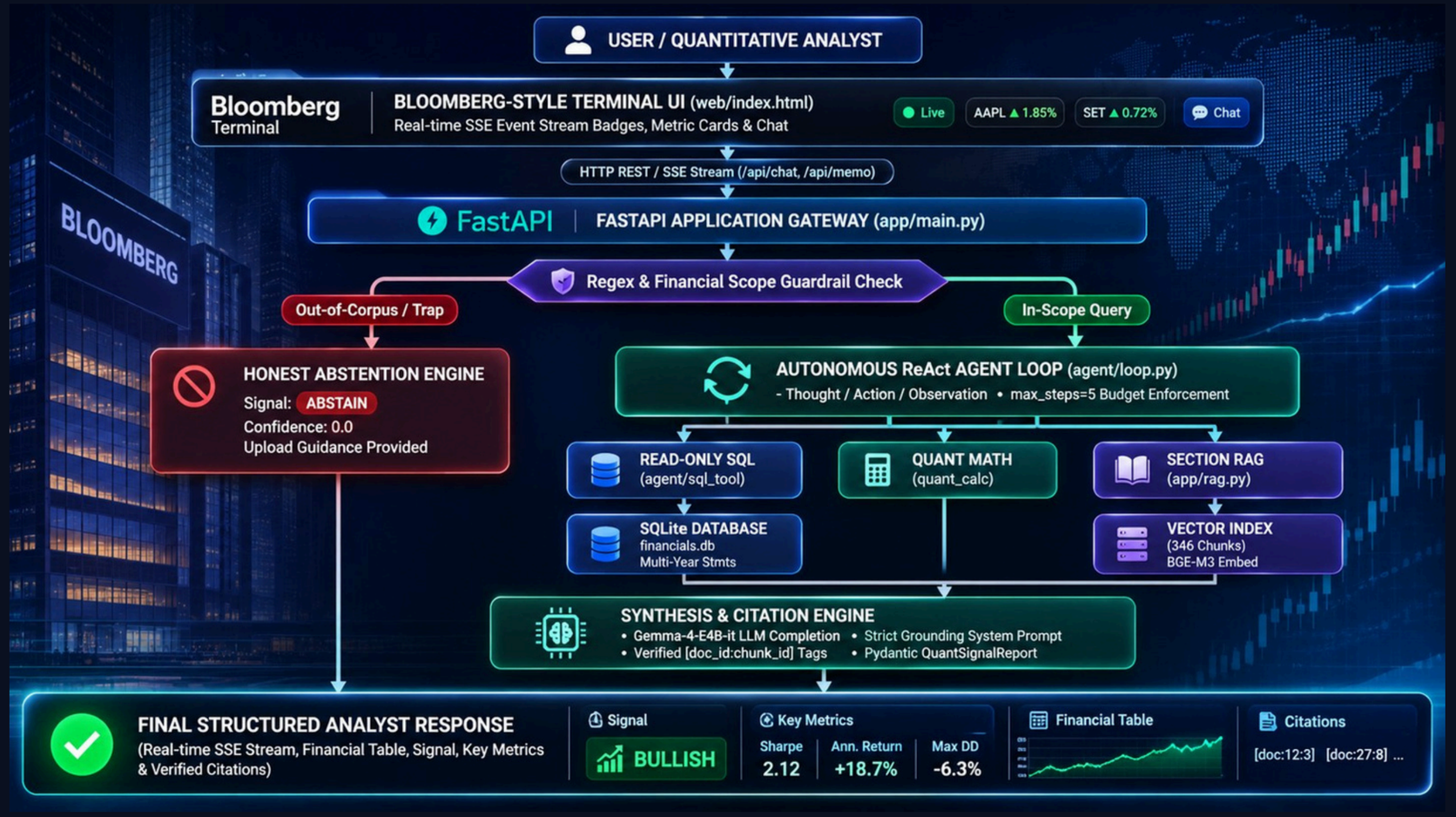


Corporate finance teams

Peer comparison with consistent accounting and paragraph-level [doc_id:chunk_id] citations.

System Architecture

A guardrail routes each query: abstain, or run a ReAct loop over three tools



1 Guardrail first

Regex and scope check sends out-of-corpus queries straight to ABSTAIN.

2 ReAct loop

Thought, action, observation with a max_steps=5 budget.

3 Cited output

Pydantic-validated report with verified citation tags.

Course Techniques Used

All 6 capability families from Lectures 1-15, each chosen for a reason

1 Prompting & structured output

Pydantic reports plus a repair parser keep LLM output machine-readable.

2 Backend & streaming

FastAPI + SSE show every tool call live, so each step is auditable.

3 Vector storage & RAG

Section-aware chunks, MIN_SCORE 0.30 and verified citation tags.

4 Rigorous evaluation

30 gold questions, 6 metrics, 2 baselines and an ablation grid.

5 Autonomous agent loop

ReAct with a 5-step budget; multi-ticker queries run per company.

6 Sandboxed SQL / code

Read-only SQLite and deterministic Python replace LLM arithmetic.

Data & Deterministic Engine

Real filings in, exact formulas out, no synthetic data

14
SEC 10-Q filings

5
Companies

346
Vector chunks

16
Tests passing

BE · GOOG · JPM · PENG · SPCX

Computed in Python, never by the LLM

DuPont ROE
Net income / Revenue × Revenue / Assets × Assets / Equity

Altman Z
1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 0.999X5 (Safe > 2.99, Distress < 1.81)

Free cash flow
Operating cash flow − CapEx

SQL guardrail
SELECT / WITH only; INSERT, UPDATE, DELETE, DROP blocked

Live Demo: Two Scenarios

One query it can answer, one it must refuse

SUCCESS CASE

“Compare BE and SPCX 2026 performance: DuPont ROE, full-year guidance and key commercial drivers, with citations.”

- query_financial_db
- calculate_quant_metrics
- search_filings_rag

Streams tool badges live, then a cited comparison, e.g. [BE_10-Q_2026_Q3:c1].

TRAP CASE

“What is PTT Public Company Limited and its Bitcoin treasury allocation guidance for 2028?”

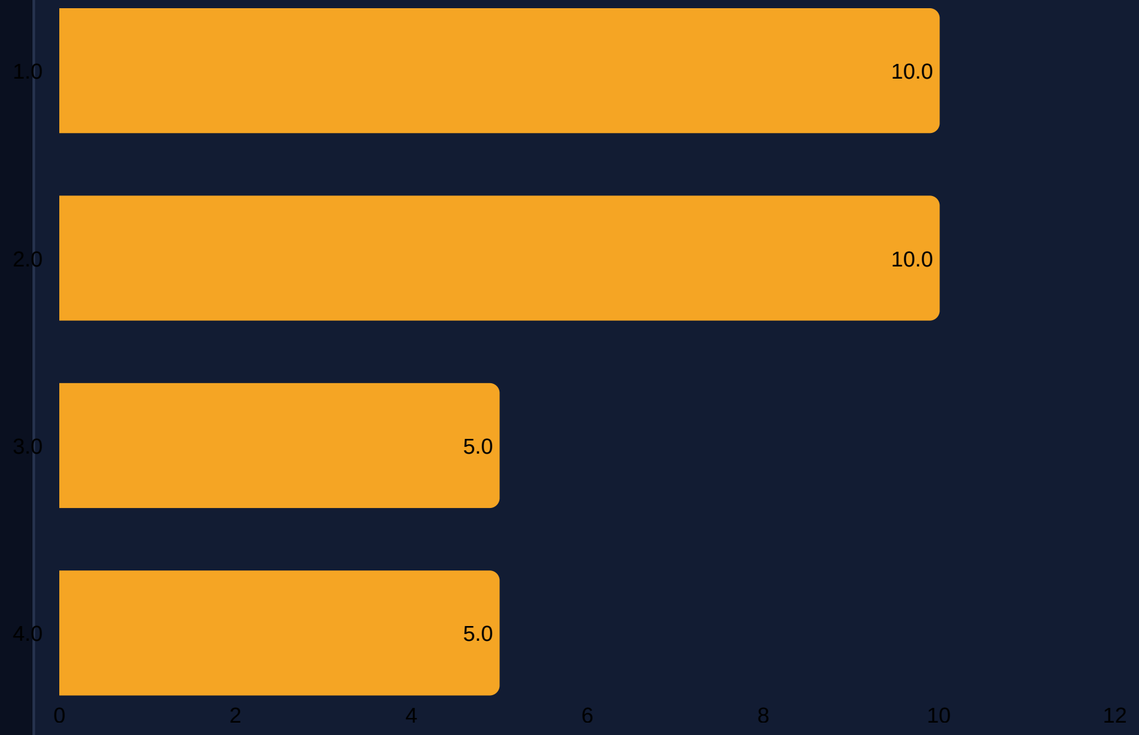
- Signal: ABSTAIN
- Confidence: 0.0

Refuses in under 50 ms and explains how to upload the missing filing.

Evaluation Design

30 gold questions, 6 metrics, 2 baselines

Test set: eval/qrels.jsonl



Metrics

Recall@3 · MRR@3 · Faithfulness (LLM judge) · Numerical accuracy · Citation precision · Trap abstention rate

Baselines

B1 Zero-shot LLM (no RAG, no tools)
B2 Naive RAG: 500-char fixed chunks, min_score 0

Results: End-to-End Comparison

Tools lift accuracy, citations and abstention, but retrieval metrics fell

System	Recall@3	MRR@3	Faithful.	Num. acc.	Cite prec.	Trap abst.
B1 Zero-shot LLM	0.00%	0.00%	0.00%	25.22%	0.00%	0.00%
B2 Naive RAG	**28.00%**	**23.33%**	**78.00%**	47.41%	73.89%	0.00%
Proposed agent	10.00%	6.67%	53.33%	**69.45%**	**92.88%**	**80.00%**

What improved

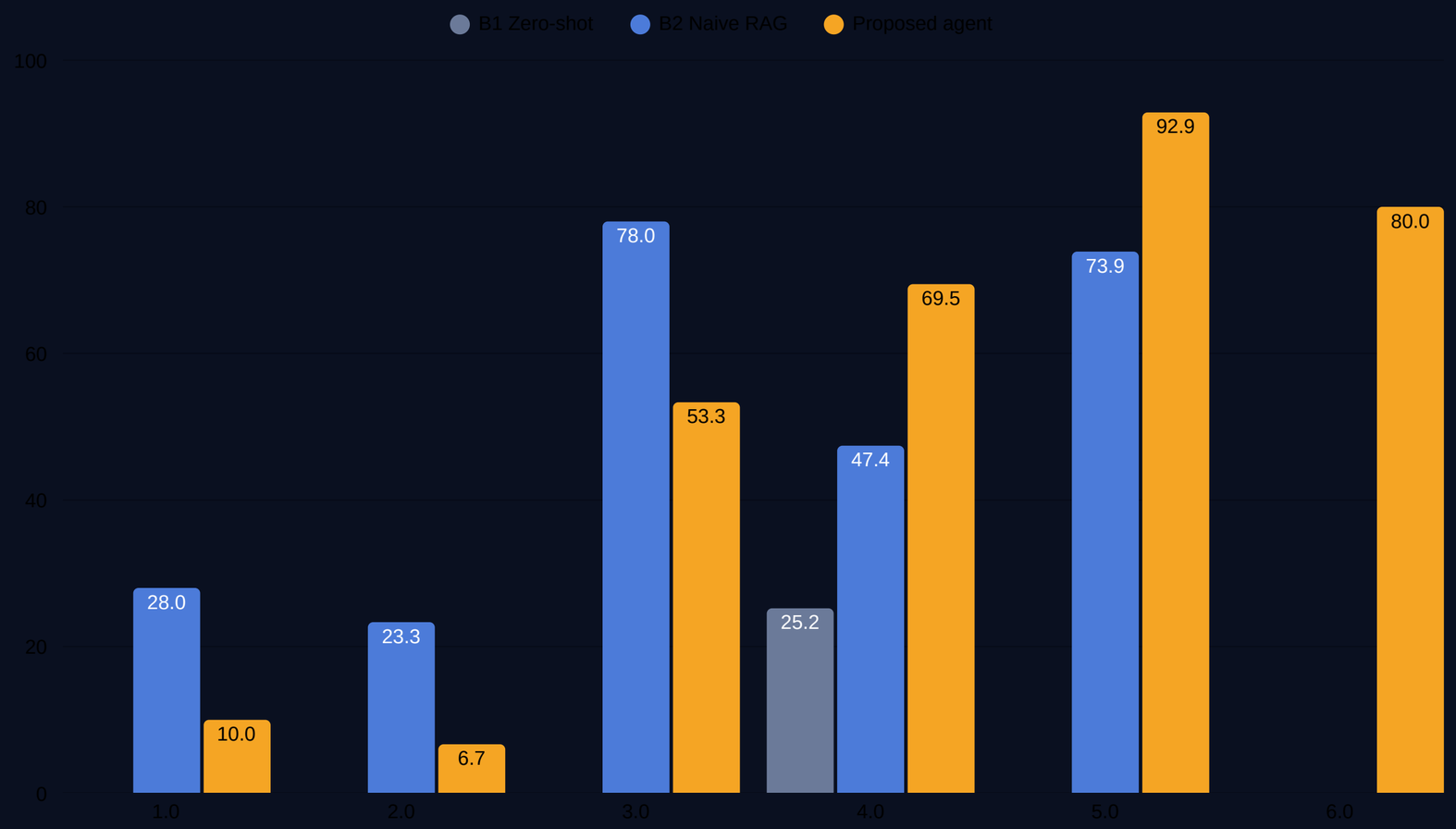
Numerical accuracy +44 pts over zero-shot; citation precision 92.88%; trap abstention 0% to 80%.

What got worse

Recall@3 10% vs 28% for naive RAG; faithfulness 53% vs 78%. Retrieval is the weak link.

Where the Agent Wins, and Where It Doesn't

The gains come from tools and guardrails, not from better retrieval



Honest reading

- Wins: exact math via SQL and quant tools; abstention via guardrails.
- Losses: retrieval and faithfulness trail naive RAG.
- Untested guess: SQL answers skip retrieval, so Recall@3 may understate the agent.

Ablation: The Threshold Trade-off

Raising min_score buys abstention but eventually destroys recall



- 0.30 is the sweet spot: 40% abstention while keeping dense footnote chunks.
- 0.50 abstains on 100% of traps but recall falls to 4% and accuracy to 24%.
- Surprise: section-aware chunking got 6% Recall@3 vs 28% for fixed chunks.

Failure Cases

Two failures found by the evaluation, both traced to a root cause

FAILURE 1 · Q18 & Q19

Period & run-rate mismatch

Query: *Compare BE and SPCX 2026 revenues and operating margins.*
What broke: Naive RAG pulled different quarters and mixed cumulative 9-month with single-quarter figures, confusing BE's Q2 $1,065.4M with its $4.4B-$4.6B full-year guidance.
Fix: Period-aligned statements in SQLite via query_financial_db.

FAILURE 2 · Q25

Over-aggressive thresholding

Query: *Across all 5 companies, state each DuPont ROE and guidance.*
What broke: A broad query diluted per-ticker similarity. Footnotes scored 0.32-0.48, so a high floor dropped 4 of 5 required sections.
Fix: MIN_SCORE=0.30 plus per-ticker tool runs in agent/loop.py.

Limitations & Next Steps

Retrieval quality is the most important thing still to fix

Limitations

- Recall@3 is only 10%; retrieval trails naive RAG
- Small test set: 30 questions, 5 companies, 10-Q only
- 1 of 5 trap queries was still answered (80% abstention)
- Filings must be clean Markdown; scanned tables unsupported
- No real-time market data; filings only

Next steps

- Rework retrieval: hybrid search and reranking, per-section thresholds
- Layout-aware vision models for image tables in PDFs
- Exchange data feeds for live price-to-earnings
- Bull vs bear sub-agent debate before the final memo
- Larger, multi-run evaluation with confidence intervals

Conclusion

Don't ask the LLM to do every calculation. Coordinate deterministic engines through grounded, accountable agent loops.

+44 pts	92.88%	80%
Numerical accuracy vs zero-shot	Citation precision	Trap abstention (baselines: 0%)

Repo: github.com/AungMoonLord/cross-market-quant-agent
Live: cross-market-quant-agent.onrender.com